

Mathematical Induction

Date_____ Period____

Write the induction proof statements P_1 , P_k , and P_{k+1} for each conjecture.

1) $P_n: 11 + 19 + 27 + \cdots + 8n + 3 = n(4n + 7)$

2) $P_n: 7^n - 4^n$ is divisible by 3

3) $P_n: 4^n \geq 4n$

Use mathematical induction to prove that each statement is true for all positive integers

4) $36 + 324 + 900 + \cdots + (12n - 6)^2 = 12n(4n^2 - 1)$